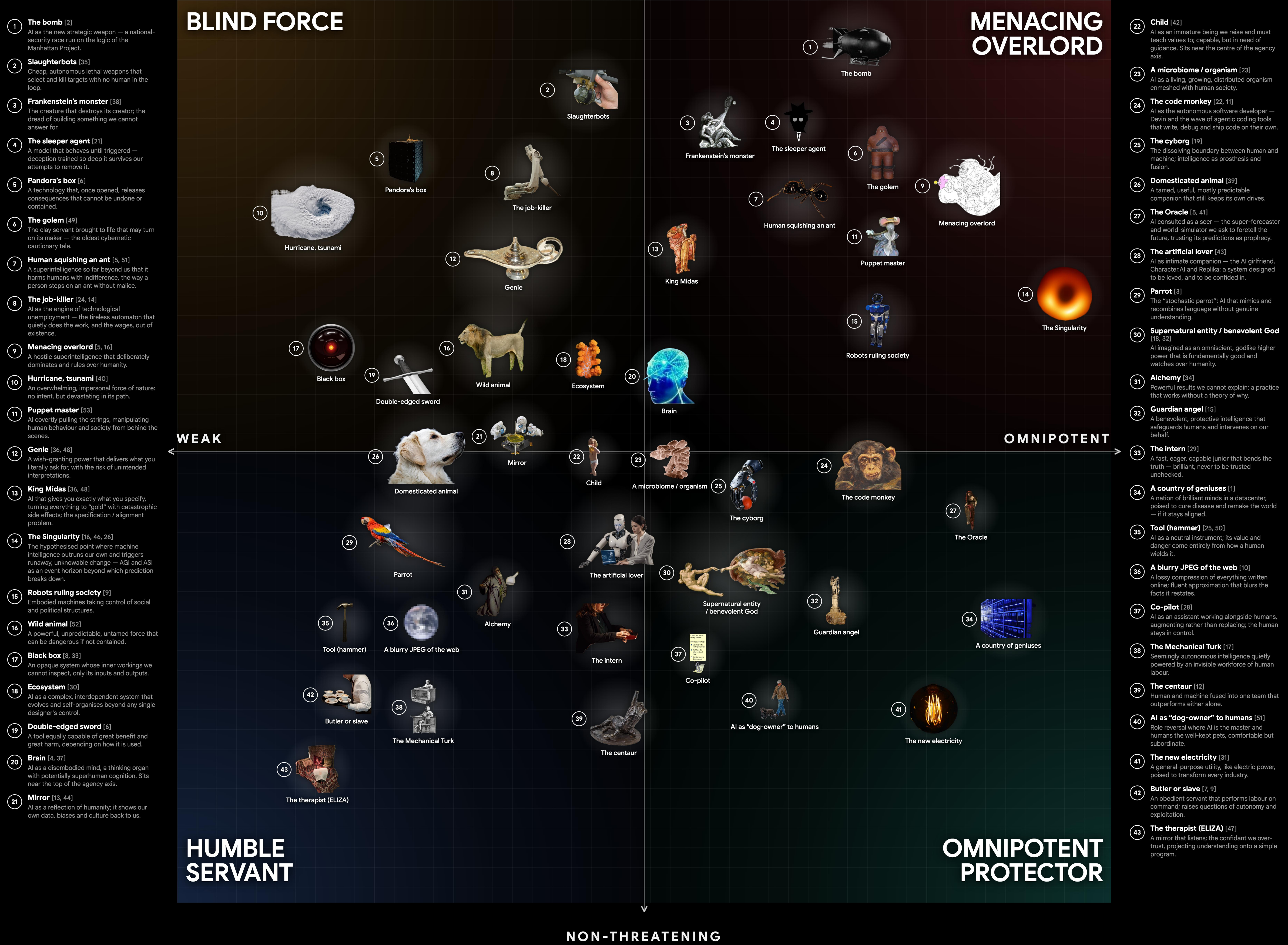


METAPHORS OF AI

clustering the metaphors by how dangerous, and how agentic, we imagine the machine to be [27, 45]



REFERENCES & FURTHER READING

The metaphors are drawn from the literature; each entry lists a source and a short link to it.

[1] Amodei, D. (2024). "Machines of Loving Grace: How AI Could Transform the World for the Better." Essay, [dariosmodel.com](#). [dariosmodel.com/machines-of-loving-grace](#)

[2] Aschenbrenner, L. (2024). *Situational Awareness: The Decade Ahead*. [situational-awareness.ai](#). [situational-awareness.ai](#)

[3] Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" *Proc. ACM FAccT*, '21, 610–623. [doi.org/10.1145/3442188.3445922](#)

[4] Berkeley, E. C. (1949). *Giant Brains, or Machines That Think*. New York: John Wiley & Sons. [archive.org](#)

[5] Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press. [global.oup.com](#)

[6] Brundage, M., et al. (2018). "The Malicious Use of Artificial Intelligence." [arXiv:1802.07228](#). [arxiv.org/abs/1802.07228](#)

[7] Bryson, J. J. (2010). "Robots Should Be Slaves." In *Close Engagements with Artificial Companions*, 63–74. John Benjamins. [benjamins.com](#)

[8] Burrell, J. (2016). "How the Machine 'Thinks': Understanding Opacity in Machine Learning Algorithms." *Big Data & Society*, 3(1). [doi.org/10.1177/2053951715622512](#)

[9] Čapek, K. (1920). *R.U.R. (Rossum's Universal Robots)*. Prague: Aventinum. [gutenberg.org/ebooks/59112](#)

[10] Chiang, T. (2023). "ChatGPT is a Blurry JPEG of the Web." *The New Yorker*, 9 Feb. [newyorker.com](#)

[11] Cognition AI (2024). "Introducing Devin, the First AI Software Engineer." [cognition.ai/blog/introducing-devin](#)

[12] Cowen, T. (2013). *Average Is Over: Powering America Beyond the Age of the Great Stagnation*. New York: Dutton. [penguinrandomhouse.com](#)

[13] Crawford, K. (2021). *Atlas of AI*. New Haven: Yale University Press. [yalebooks.yale.edu](#)

[14] Frey, C. B., & Osborne, M. A. (2017). "The Future of Employment." *Technological Forecasting & Social Change*, 114, 254–280. [doi.org/10.1016/j.techfore.2016.08.019](#)

[15] Goertzel, B. (2012). "Should Humanity Build a Global AI Nanny...?" *Journal of Consciousness Studies*, 19(1–2), 96–111. [ingentaconnect.com](#)

[16] Good, I. J. (1965). "Speculations Concerning the First Ultra-intelligent Machine." *Advances in Computers*, 6, 31–88. [sciencedirect.com](#)

[17] Gray, M. L., & Suri, S. (2019). *Ghost Work: How to Stop Silicon Valley from Building a New Global Underclass*. Boston: HMH. [ghostwork.info](#)

[18] Harari, Y. N. (2016). *Homo Deus: A Brief History of Tomorrow*. London: Harvill Secker. [yeharari.com](#)

[19] Haraway, D. (1985). "A Cyborg Manifesto." *Socialist Review*, 80, 65–108. [archive.org](#)

[20] Hicks, M. T., Humphries, J., & Slater, J. (2024). "ChatGPT is Bullshit." *Ethics and Information Technology*, 26(2), 38. [doi.org/10.1007/s10676-024-09775-5](#)

[21] Hubinger, E., et al. (2024). "Sleepers: Training Deceptive LLMs that Persist Through Safety Training." *Anthropic*. [arXiv:2401.05566](#)

[22] Jimenez, C. E., et al. (2024). "SWE-bench: Can Language Models Resolve Real-World GitHub Issues?" *ICLR*. [arxiv.org/abs/2310.06770](#)

[23] Kelly, K. (2010). *What Technology Wants*. New York: Viking. [k.k.org](#)

[24] Keynes, J. M. (1930). "Economic Possibilities for Our Grandchildren." In *Essays in Persuasion* (1931). [gutenberg.net.au](#)

[25] Kranzberg, M. (1986). "Technology and History: 'Kranzberg's Laws.'" *Technology and Culture*, 27(3), 544–560. [doi.org/10.2307/3105385](#)

[26] Kurzweil, R. (2005). *The Singularity is Near*. New York: Viking. [penguinrandomhouse.com](#)

[27] Lakoff, G., & Johnson, M. (1980). *Metaphors We Live By*. University of Chicago Press. [press.uchicago.edu](#)

[28] Licklider, J. C. R. (1960). "Man-Computer Symbiosis." *IRE Trans. Human Factors in Electronics*, HFE-1, 4–11. [ieeexplore.ieee.org](#)

[29] Molick, E. (2024). *Co-Intelligence: Living and Working with AI*. New York: Portfolio. [penguinrandomhouse.com](#)

[30] Nardi, B. A., & O'Day, V. L. (1999). *Information Ecologies: Using Technology with Heart*. MIT Press. [mitpress.mit.edu](#)

[31] Ng, A. (2017). "Artificial Intelligence is the New Electricity." Keynote, Stanford GSB. [gab.stanford.edu](#)

[32] O'Gieblyn, M. (2021). *God, Human, Animal, Machine*. New York: Doubleday. [penguinrandomhouse.com](#)

[33] Pasquale, F. (2015). *The Black Box Society*. Cambridge, MA: Harvard University Press. [hup.harvard.edu](#)

[34] Rahimi, A. (2017). "Machine Learning Has Become Alchemy." *NeurIPS Test-of-Time Award lecture*. [nips.cc](#)

[35] Russell, S., et al. (2017). *Slaughterbots* [short film]. Future of Life Institute. [autonomousweapons.org](#)

[36] Russell, S. (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. New York: Viking. [humancompatible.ai](#)

[37] Shanahan, M. (2023). "Talking About Large Language Models." [arXiv:2212.03551](#) (Comm. ACM, 2024). [arxiv.org/abs/2212.03551](#)

[38] Shelley, M. (1818). *Frankenstein; or, The Modern Prometheus*. London: Lackington et al. [gutenberg.org/books/84](#)

[39] Silverstone, R., & Haddon, L. (1996). "Design and the Domestication of ICTs." In *Communication by Design*. Oxford UP. [oup.com](#)

[40] Suleyman, M., with Bhaskar, M. (2023). *The Coming Wave*. New York: Crown. [the-coming-wave.com](#)

[41] Tetlock, P. E., & Gardner, D. (2016). *Superforecasting: The Art and Science of Prediction*. New York: Crown. [penguinrandomhouse.com](#)

[42] Turing, A. M. (1950). "Computing Machinery and Intelligence." *Mind*, 59(236), 433–460. [doi.org/10.1093/mind/LIX.236.433](#)

[43] Turkle, S. (2011). *Alone Together*. New York: Basic Books. [sherryturkle.com](#)

[44] Vallor, S. (2024). *The AI Mirror*. Oxford University Press. [global.oup.com](#)

[45] Vickers, B., & Allado-McDowell, K. (eds.) (2020). *Atlas of Anomalous AI*. London: Ignota Books. [ignota.org](#)

[46] Vinge, V. (1993). "The Coming Technological Singularity." *VISION-21 Symposium*, NASA. [edoras.sdsu.edu/~vinge](#)

[47] Weizenbaum, J. (1976). *Computer Power and Human Reason*. San Francisco: W. H. Freeman. [archive.org](#)

[48] Wiener, N. (1960). "Some Moral and Technical Consequences of Automation." *Science*, 131(3410), 1355–1368. [doi.org/10.1126/science.131.3410.1355](#)

[49] Wiener, N. (1964). *God & Golem, Inc.* Cambridge, MA: MIT Press. [mitpress.mit.edu](#)

[50] Winner, L. (1980). "Do Artifacts Have Politics?" *Daedalus*, 109(1), 121–136. [jstor.org](#)

[51] Wozniak, S. (2015). Remarks at the Freescale Technology Forum (reported in Time, June 2015). [time.com](#)

[52] Yampolekij, R. V. (2020). "On Controllability of Artificial Intelligence." [arXiv:2008.04071](#). [arxiv.org/abs/2008.04071](#)

[53] Zuboff, S. (2019). *The Age of Surveillance Capitalism*. New York: PublicAffairs. [publicaffairsbooks.com](#)